Analytical Cume
March 11, 2011

(Note… equations at end of cume)

Separations

1. (30 points) Separation of a commercially-synthesized epoxy resin (polymer) on a size-exclusion column with a porous silica packing produced peaks in a chromatogram with the following retention times. The column is 6.2 mm diameter x 250 mm long, with a mobile phase of tetrahydrofuran and flow rate of 1.3 mL/min. The solid portion of the particles occupies 3.0% of the total volume of the column, the volume within the pores occupies 26.8%, and the volume between particles occupies 70.2%. Assume that this is an ideal exclusion column (there is no adsorption). The part of the resin oligomers that do not include the repeating unit (i.e. the end groups) add up to MW of 591. Polymers containing repeating units of 264 for n = 0, 1, 2, 3, etc. are possible. Below is a table of the results of a separation of a mixture of polymer oligomers. (Note: MW = molecular weight or molecular mass).

Species (MW = 591 + n x 264)	Retention Time / min
n=0	5.65
n=1	5.24
n=2	4.90
n=3	4.71
n=4	4.51
n=5	4.35
n=6	4.20

(a) (10 points) What is the quantitative relationship of retention volume to molecular weight (or molecular mass) to separate molecules on a size exclusion column?

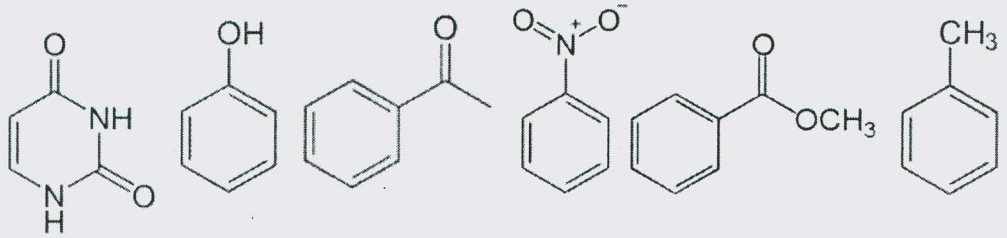
(b) (10 points) At what time would you expect the n=7 oligomer to elute? Show your calculation.

(c) (10 points) Would you be able to resolve the peak for the n=7 oligomer? Why or why not? Use calculations to support your argument.

2. (25 points) The following table provides a summary of the retention times of several chemical species eluted from partition chromatography of a mixture on a HPLC column containing 5-μm diameter particles. The length of the column is 20.0 cm, its diameter is 0.5 cm, the velocity of the mobile phase is 1.0 mL/min, volume of the stationary phase (V_s) is 1.79 mL, and the volume of the mobile phase (V_m) is 2.14 mL. The following data pertain to the separation (note W = width of peak at baseline) and t_R is retention time):

species	t_R, min	W, min
non-retained	0.8	---
uracil	2.5	0.10
phenol	4.6	0.20
acetophenone	6.4	0.50
nitrobenzene	8.0	0.60
methyl benzoate	11.6	0.70
toluene	22.5	1.10

Structures of the chemical species are shown below.



Uracil, phenol, acetophenone, nitrobenzene, methyl benzoate, toluene

(a) (5 points) Calculate the number of plates for uracil.
(b) (5 points) Calculate the resolution for uracil and phenol.
(c) (10 points) Calculate the length of the column to separate uracil and phenol so that they are just barely baseline resolved at $R_s = 1.5$.
(d) (5 points) What would be the time required to separate uracil on this shorter column (length calculated in part (c))?

3. (45 points) Basics
(a) (10 points) What does GC stand for? Describe what kinds of mixtures of molecules is it most suitable for? What makes the mobile phase move through the column? Give one example of a column and one example of a detector.
(b) (10 points) What does HPLC stand for? What kinds of mixtures of molecules is it most suitable for? What makes the mobile phase move through the column? Give one example of a column and one example of a detector.
(c) (10 points) What does UPLC stand for? What kinds of mixtures of molecules is it most suitable for? What makes the mobile phase move through the column? Give one example of a column and one example of a detector.
(d) (15 points) What are some advantages of UPLC over HPLC? What are the greater challenges of UPLC compared to HPLC?

Equations that Might or Might Not be Helpful on Analytical Cume

$u = L/t_m$

$v = L/t_R$

v = u • (moles of solute in mobile phase/total moles solute)

$$\sigma^2 = H\,x$$

$$\sigma^2 = H\,L$$

$$N = \frac{L}{H}$$

$$N = \frac{L}{H} = \frac{L^2}{\sigma^2} = \frac{16\,L^2}{w^2}$$

$$N = \frac{16\,t_r^{\,2}}{w^2} = \frac{5.55\,t_r^{\,2}}{w_{1/2}^{\,2}}$$

$$H = A + \frac{B}{u} + C_{SP}\,u + C_{MP}\,u$$

$K = C_s/C_M$

k' = (time solute spends in the stationary phase / time solute spends in the mobile phase)

$$k'_A = \frac{moles_s}{moles_M} = \frac{C_s V_s}{C_M V_M} = K_A \frac{V_s}{V_M} = \frac{(t_{r,A} - t_M)}{t_M}$$

$$\alpha = \frac{t_{r,B} - t_m}{t_{r,A} - t_m} = \frac{t'_{r,B}}{t'_{r,A}} = \frac{k'_B}{k'_A} = \frac{K_B}{K_A} \quad \text{where } t'_{r,B} > t'_{r,A}$$

$$R_S = \frac{2\left(t_{r,B} - t_{r,A}\right)}{w_A + w_B} = \frac{\Delta t_r}{w_{av}}$$

$$R_s = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{k'_b}{1 + k'_{av}}\right)$$

$$t_{r,B} = \frac{16\,R_s^2\,H}{u}\left(\frac{\alpha}{\alpha - 1}\right)^2\left(\frac{\left(1 + k'_b\right)^3}{\left(k'_b\right)^2}\right)$$